

BIKRANT NATH

bikrantnath4@gmail.com | +91 9101121475 | GitHub | Portfolio | LinkedIn | Open to Relocation

TECHNICAL SKILLS

Languages & Fundamentals: Python, JavaScript, SQL, Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems

AI, Generative AI & Agents: Machine Learning, LLM API Integration, LangGraph, LangChain, RAG, Prompt Engineering, MCP, Vector DB, Model Evaluation, NLP, GANs, PyTorch, Hugging Face Transformers, Scikit-learn

Product & Full-Stack Development: Fast API, Node.js, Express.js, React, REST APIs, MongoDB, MySQL

MLOps, Cloud & Infra: CI/CD (Github Action), Docker, Kubernetes, MLflow, DVC, AWS, GCP, Linux, Nginx, Git, Jira, Pandas, NumPy

EXPERIENCE

Founder & Full-Stack Developer

https://github.com/bikrantnath22/athaana_ | 2022

Athaana - Grocery eCommerce Startup

Nalbari, Assam

- Created Athaana, a full-stack grocery eCommerce platform from scratch (product listings, cart, checkout, JWT authentication, live order tracking) with React, Node.js, Express, MongoDB, and CSS, hosted on DigitalOcean.
- Led a 4-person founding team (2 non-technical co-founders) to launch and run the platform in Nalbari, processing 250+ real customer orders via a live order-management and tracking system.
- Built the venture from 0 to 1 — product, customer onboarding, operations, and all engineering — before closing the business

Software Development Intern

Jan 2022 – Jul 2022

Fineworks Media & Solution Technologies

Guwahati, India

- Developed, tested and documented REST APIs for a full-stack booking platform (React, node.js/Express, MongoDB) that included end to end confirmation and availability workflows.
- Worked with the development team on debugging and structured code reviews to increase the reliability of the production transactions.

PROJECTS

AI Research & Fact-Verification Agent | LIVE : <https://factverification.duckdns.org/>

<https://github.com/bikrantnath22/AI-Research-Fact-Verification-Agent> | 2026

- Developed a 4-agent (Planner, Retriever, Synthesiser, Verifier) LangGraph pipeline with Qdrant vector search, live Tavily web search and inline citations, reducing hallucinated answers with weighted 3-signal risk model (semantic entropy, ensemble disagreement, faithfulness).
- Enabled reuse of zero-custom-integration for any MCP-compatible client, as measured by 3 independently callable tool servers, by exposing retrieval, web search, and verification as separate MCP servers over Streamable HTTP.
- Executed unit testing and integration test suites for a 5-service Docker Compose stack (FastAPI backend) deployed via GitHub Actions CI/CD to AWS EC2 behind an Nginx reverse proxy to verify pipeline reliability.

Tech Stack: Python, LangGraph, FastAPI, Groq, Qdrant, MCP, Hugging Face, DeBERTa, Docker Compose, GitHub Actions, AWS EC2, Nginx

RAG Guardrail System | <https://github.com/bikrantnath22/RAG-Guardrail-System> | 2026

- Developed a 4-layer input guardrail (regex heuristics, DeBERTa prompt-injection classifier, MiniLM semantic similarity, LLM-as-judge) to avoid prompt injection and jailbreaks from passing into the RAG pipeline.
- Built a 2-layer output guardrail (role-aware PII redaction + LLM sensitivity check across 4 data tiers) and 3-tier RBAC (Guest/Employee/Admin) with toggleable dashboard, containerized via Docker, to visualize and block data exfiltration across 3 real attack scenarios.

Tech Stack: Python, FastAPI, DeBERTa, Sentence-Transformers, Hugging Face, Groq, PyTorch, Docker

DP-CTGAN with Rare Minority Sampling | (M.Tech Research)

<https://github.com/bikrantnath22/dp-ctgan> | 2025–2026

- Achieved strong minority-class fidelity at $\epsilon = 3$ - 0.7933 AUPRC/0.7512 F1 (Credit Card Fraud), 0.6460/0.6495 (Adult Census), 0.4576/0.4973 (Taiwan), 0.3441/0.3853 (MIMIC-III) — with negligible MIA leakage, using a DP-CTGAN (PyTorch, Opacus DP-SGD, Rényi accountant) with a normalizing-flow rare-class sampler and PrivacyGate filter.
- Verified fidelity and privacy rigor via KS/JS distribution metrics and an unbiased shadow-model MIA with PCA-derived pseudo-labels, catching and correcting a hidden AUC-inflation artifact in the evaluation pipeline.
- The quality of the model is preserved during retraining cycles, tracking AUPRC/Precision/Recall/F1 per cycle in an end-to-end MLOps pipeline (Data → Features → Training → Monitoring → Retraining) on all 4 datasets.

Tech Stack: Python, CTGAN, PyTorch, Opacus, NumPy, Pandas, Scikit-learn, Optuna

EDUCATION

M.Tech – Computer Science & Engineering, Tezpur University

Jul 2026 | CGPA: 7.63/10

MCA – Master of Computer Applications, Tripura University

Jul 2024 | CGPA: 7.71/10

CERTIFICATIONS & ACHIEVEMENT

NPTEL: Machine Learning, Cloud Computing, IoT, Ethical Hacking [Link](#)

Codestellation Hackathon: 1st Runner-Up 2022 (Assam Engineering College)